Intermediate Submission 1 Group 58

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1 INTERMEDIATE SUBMISSION 1

1.1 Project Idea

We aim to develop a 2D educational game inspired by CodeMonkey and Scratch, which is designed to teach programming basics interactively. Players control a character across a grid-based map using simple Java code. The game's goal is to navigate the character from a start to an endpoint while avoiding obstacles or level-specific constraints. The user executes the in-game commands in a sandbox environment, and user progress is saved in properties files. Visual feedback, simple tutorials, and documentation are provided to assist the user.

1.2 Use-Case Diagram

This diagram shows the primary functionalities of the game system. The main interaction, 'Start New Level', and supporting features all improve the functionality of the game and the user's experience, such as 'Access Help/Hints', 'Select Level', etc.

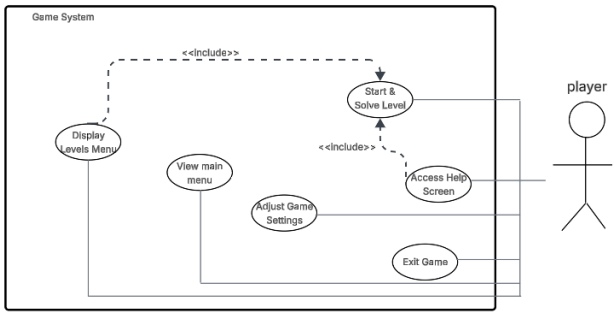

Figure 1: Use Case Diagram

1.3 Use-Case Description

Section	Content
Identifier	Level-01
Name	Start & Solve Level
Authors	Sofia Constantinou, Nicole Almeida, Alex Christou
Priority	Must
Summary	User starts the game, and tries to complete the level. Based on his actions and instructions submitted, the respective result will be followed
Actors	User of the game
Trigger	User exits game or level, or never provides a correct solution
Pre-conditions	-
Post-conditions	Level is completed
Result	User completing the respective level
Main Scenario	1. The user starts the game 2. The system displays the next level within a gamified environment 3. The student submits code to solve the loaded puzzle 4. The user runs the code and the result is shown visually 5. The system provides feedback about whether the solution is correct 6. The user receives the respective amount of stars 7. The user completes the level
Alternative Scenarios	3a. The user does not submit any instructions, instead submits an empty log 6a. The user completes the program after a failed attempt 6b. The user provides a correct solution which is not optimal, and does not receive 3 stars
Exception Scenarios	Trigger: The user exits the level Trigger: The user never ends up providing a correct solution Trigger: The user exits the game
Requirements	F-1, F-2, F-4, F-5, F-10, F-13

Figure 2: Main Use Case Description

1.4 Requirements

1.4.1 Functional Requirements:

(1) The system must display the levels, distinguishing between those that are unlocked and those that are locked.
(2) The system must provide the user with the ability to start playing a level once it has been selected.
(3) The system must have background music and in-game sounds.
(4) The system must provide the user with the ability to adjust the volume.
(5) The system must provide the user with the ability to select their avatar from a predefined set.
(6) The system must have a settings button, which leads to a view with options to adjust the sound volume, select an avatar, exit the settings, and return to the previous screen.
(7) The system must allow the user to exit the game through an exit button located in the main menu.
(8) The system must include a restart functionality, which restores the grid of the level and the sprite to its initial position.
(9) The system must include an in-game menu containing the buttons to reset a level, execute the code, stop the execution, show the help screen, and display the in-game settings.
(10) The system must display an appropriate message and give the chance to retry the level if the user fails to complete it.
(11) The system must provide the user with the capabilities to run in-game Java commands that allow direct interaction with the sprite in a given level.
(12) The system must provide the player with the capability to view in-game Java documentation and relevant game and level information.

(13) The system must allow the users to view the full set of in-game Java commands in the help screen.
(14) The system must render a 2D grid-based game environment with a movable character (sprite), static obstacles, a start, and an end goal for a given level.
(15) The system must move the sprite step-by-step based on the instructions.
(16) The system must display an appropriate message, including the received stars, once the character completes the level, and give the option to restart, move to the next level, or go to the levels menu.
(17) The system must calculate the efficiency of a solution by ranking it out of 3 stars (1-3) in comparison to the ideal solution.
(18) The system must detect collisions with obstacles during executions, leading to the failure of a level.
(19) The system must provide the user with the ability to move on to the next level once a level is completed, by pressing the next level button.
(20) The system must include a help screen in the main menu, containing basic information regarding how the game works.
(21) The system could include levels that require the user to collect a key, open a door, and go through it to pass the level.

1.4.2 Quality Requirements:

(1) The system's settings and game progress must be saved and the modifications must be applied to the property file within 1 second of the user changing them.
(2) The system must prevent unintended command execution via a safe sandboxed environment and log the error in the in-game level console within 1 second after the in-game level code finishes the execution.
(3) The system must support a minimum screen resolution of 1024x768 pixels.
(4) The system must load any selected level, including all its assets, and be ready for user interaction within 2 seconds of selecting the level.

1.4.3 Constraint Requirements:

(1) The system must be compatible with JDK 21 and JavaFX.
(2) Game progress and settings must be persisted using properties files, with a maximum file size of 5MB.
(3) Each system level can have up to 4MB of uncompressed asset data.
(4) The system's grid of tiles must have a size of at least 4x4 tiles and not exceed 16x16 tiles.